A2 UI/UX Research Report – Fake Uber Final Design

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1. Introduction

Abstract. This report evaluates the final interface of *Fake Uber*, a proof-of-concept car rental app built for COS30017 Assignment 2. The final design has 14 screens covering browsing, searching, sorting, favourites, renting, driver documents, errors, notifications and dark mode. Each design decision is examined against published usability research, Android platform guidance and the Web Content Accessibility Guidelines (WCAG). The design satisfies every functional requirement of the brief and holds up well against Nielsen's heuristics. Its remaining weaknesses are almost all about accessibility: six light-theme colour pairs and the dark-theme error red fall below WCAG contrast thresholds, and several controls are drawn smaller than the 48dp touch target Android recommends. These can be fixed when the app is built, without changing the layout.

The brief asks for a "simple proof-of-concept app" in which clients browse five cars one at a time, rent one for up to seven days with a 500-credit balance, and see the result on the first screen. The unit's tutor stressed that the logic should stay simple and that the interface is where the effort should go. This report therefore asks one question: **does each part of the Fake Uber interface help a first-time user rent a car quickly and without mistakes, and what evidence supports that?**

The report follows the user's path through the app. It starts with the method and the visual language, then moves from browsing on the Home screen to finding cars, renting, uploading driver documents, recovering from errors and switching to dark mode. It ends with an accessibility audit and an honest account of what a proof-of-concept cannot prove.

2. Context and method

The design was shaped by the brief's constraints more than by aesthetic preference. Cars must appear one at a time with a Next button, data lives only in memory, every rental is capped at 400 credits, Back must count as a cancellation, and the app must show a Toast or Snackbar whether a booking is saved or not. A Search bar, a Sort menu with three orders, a Favourites section, a Dark Mode switch and at least two non-TextView widgets are also required. Any screen that ignored one of these would fail the brief, however attractive it looked.

Three free and widely used methods were combined to evaluate the design. The first is a **heuristic evaluation** against Jakob Nielsen's ten usability heuristics (Nielsen, Princeton COS436 handout), a widely taught expert-review method in HCI. The second is a **guideline review** against Android's own documentation and the Material Components library that the app will be built with, because a design that cannot be built faithfully is not useful. The third is an **accessibility measurement**: contrast ratios were calculated with the WCAG relative-luminance formula from the design's colour tokens, and touch targets were compared with Android's 48 × 48dp recommendation (Android Developers, Make apps more accessible).

The sources were chosen to be free to read and to carry weight. Most are Nielsen Norman Group articles, which summarise decades of usability testing; official Android and W3C documentation; and peer-reviewed HCI research where it is openly available, such as Parhi, Karlson and Bederson (2006) on thumb target sizes. Paywalled books were avoided so that every claim in this report can be checked by the reader.

The method has one clear limit. No usability test with real participants was run, so the findings describe likely problems, not observed ones. Section 11 returns to this point.

3. Visual language

Fake Uber uses a deliberately small palette with three jobs: warm cream (#F4F0E8) for the background, near-black (#15171B) for information that must always be found, and orange (#F97316) for the single action the user should take next. Near-black carries the credit balance, the Next button and the Back/Save bar. Orange is reserved for Rent, Save, the slider and the favourite heart. Because orange is rare, the eye lands on the primary action without searching. This follows the Laws of UX advice for Hick's law to "highlight recommended options" so users are not overwhelmed by equal-looking choices (Laws of UX, Hick's Law), and Nielsen's eighth heuristic, aesthetic and minimalist design.

Layout relies on spacing and cards rather than lines. Each part of the Home screen (search, Favourites, Available now, My rentals) is a titled block with clear space above it, and each fact on the car card sits inside its own rounded chip. Nielsen Norman Group describes proximity as one of the strongest grouping cues, strong enough to override colour or shape, and advises placing headings closer to the content they introduce than to the section before (Harley, 2020). The design does exactly this, so a user can tell which numbers belong to which car at a glance.

The typeface is Inter throughout, a sans-serif designed for screens with a tall x-height that keeps small labels readable (Google Fonts, Inter). Hierarchy comes from size and weight: a 25px extra-bold headline, 16–18px bold section titles, 13–14px body text and 11–12px labels. Corners are large (18–26px) and buttons are pill-shaped, which gives the app a friendly, consumer feel close to Material Design 3's larger shape scale.

Every colour is defined as a named token (bg, surface, text, text-2 and so on) with a light value and a dark value. This is not just tidy design practice. Android's dark theme guide warns against hard-coded colours and asks developers to use theme attributes so the app can switch themes cleanly (Android Developers, Dark theme). The tokens map directly onto `values/colors.xml` and `values-night/colors.xml`, and they also satisfy the brief's request for at least two styles used in at least two places.

4. Browsing on the Home screen

Showing one car at a time is the brief's choice, not the designer's, and it is a risky pattern. It is effectively a carousel, and Nielsen Norman Group's research on mobile carousels found that sequential access hides content: users should be able to reach the last item in "3–4 steps", and small dots are weak, easily missed position cues (Budiu, 2018). Fake Uber stays inside these limits. With five cars, the last one is four taps of Next away. Instead of dots, the design shows a labelled counter ("1 / 5") beside the *Available now* title, which tells the user both where they are and how much is left. The counter also changes meaning correctly: it drops to "1 / 4" after a rental, "1 / 1" after a narrow search and "0 / 0" when nothing matches. This is Nielsen's first heuristic, visibility of system status, applied to the one place where the brief makes status hardest to see.

The car card answers the brief's rule that details appear under each image. It reads top to bottom in the order people judge a rental: what it is (type tag, brand, model), what it costs (a large "60 credits / day"), whether it is good (star rating and "4.5 / 5"), and the facts that separate similar cars (year, kilometres, daily cost). The numbers are shown, not hidden behind a tap, which favours recognition over recall: people compare what they can see more easily than what they must remember from the previous card (Budiu, 2024). One small redundancy remains: the price appears twice, in the headline figure and again in the *Daily cost* chip. Replacing that chip with a fact not shown elsewhere, such as fuel type, would give the card more information in the same space.

The two actions sit at the bottom of the card in the area that a thumb reaches most easily. Steven Hoober's field observations found that most people hold their phone in one hand and tap with the thumb (Hoober, 2013). Rent is wide and orange because it is the goal; Next is narrow and black because it is the most frequent action but not the final one. Both are 52dp tall. Parhi, Karlson and Bederson's MobileHCI study recommends at least 9.2mm for single thumb targets (Parhi et al., 2006), and 52dp is about 8.3mm on a typical phone, close to that figure and above Android's 48dp minimum.

When a search leaves a single car, the Next button turns grey. The "1 / 1" counter right above it explains why, so the disabled state is understandable rather than mysterious.

5. Finding and organising cars

Search is always visible at the top of Home rather than hidden behind an icon, because finding a car by name is one of the brief's core tasks. With only five cars, results can filter as the user types, the pattern Theresa Neil and Rachel Malley recommend for small data sets (Neil and Malley, 2012). While a query is active, the field gains an orange outline and a clear (×) button, so the user can see that the list is filtered and undo it in one tap.

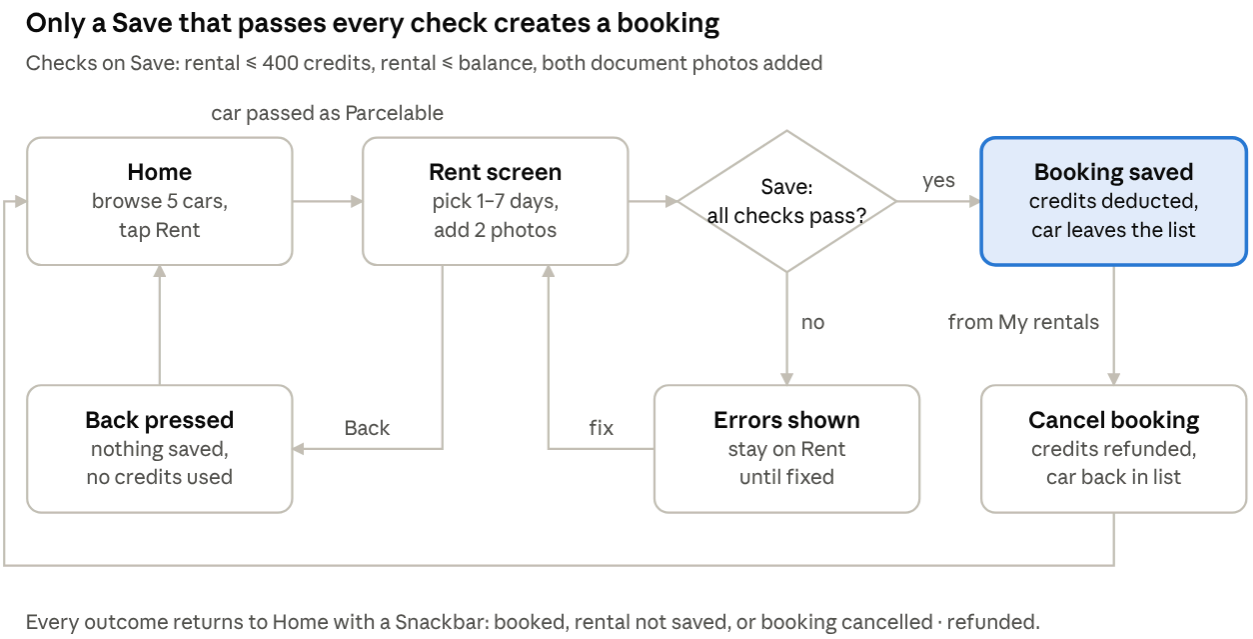
The final design adds a screen many student projects forget: what happens when nothing matches. Instead of an empty card, the user sees a large search icon, the message *No cars match "xyz"*, a hint to try a different name or model, and a **Clear search** button. Kate Kaplan's guidelines for empty states ask for exactly three things: communicate system status, provide learning cues, and offer a direct pathway forward (Kaplan, 2021). The screen meets all three, and echoing the user's own query makes it clear that the app understood the input and simply found no car.

Sorting uses a labelled button that opens a small menu titled *Sort by* with the brief's three orders: rating high to low, year newest first, and cost lowest first. Each option states both the field and the direction, so users do not have to guess what "Year" means. The current choice is highlighted and ticked. A menu is a recognition-based control: users scan visible options rather than remember them (Budiu, 2024), and with only three items Hick's law predicts a near-instant choice. One improvement remains: after the menu closes, nothing on Home shows the active order. A short "Sorted by rating" line above the car card would keep that status visible.

Favourites can be added in two ways, as the brief allows: tapping the heart on the photo or long-pressing the photo. The heart is the primary route because it is always visible; long-press is offered as a shortcut, and the empty Favourites area teaches it in one line ("Tap ♡ or long-press a car photo to save it here"). This matters because hidden gestures are hard to discover: the GhostUI study presented at CHI 2026 documents how mobile apps conceal interactions such as long-press from users (Kweon et al., 2026). Saved cars appear as chips with a thumbnail, name and price, and a counter ("2 saved") summarises the section. The filled versus outlined heart signals the state by shape, not colour alone.

6. The renting flow

The Rent screen is the strongest part of the design because it turns a rule-heavy task into a sequence of small, visible decisions. Tapping Rent opens a second screen that repeats the photo, the meta line ("Brand · 2025 · Electric"), the price and the rating, so the user never has to go back to check which car they chose. Three spec cards (seats, gearbox, fuel) add the details a driver needs before committing. Below them, the screen is split into three cards in the order a renter thinks: *How many days?*, *Driver details* and *Credits*.



Renting flow · Home, Rent, one decision, four outcomes

The flow has only one door into a booking: a Save that passes all three checks. Every other path, whether Back, an error or a later cancellation, leaves the user's credits where they started.

The number of days is chosen with a discrete slider from 1 to 7, with each step labelled underneath and the selected number shown large in the card header. A discrete slider cannot produce an invalid value such as 0 or 8 days, which is Nielsen's fifth heuristic, error prevention, built into the control itself. The Material Components documentation describes discrete sliders as the right choice when the user needs a precise value, and strongly encourages a content description for screen readers (Material Components for Android, Slider). The slider also counts as one of the brief's non-TextView widgets, alongside the RatingBar and the dark-mode Switch. The helper line "Pick-up is right now. Return within a week." quietly explains the brief's rule that there are no future bookings.

The Credits card is where the design does its most useful work. It shows the calculation line by line ("55 cr × 3 days = 165 cr"), the current balance, the balance after rental, and a *Rental limit* bar that fills towards 400 credits. Every value updates as the slider moves. The user can therefore see that a rental is too expensive before pressing Save, rather than discovering it through an error. Hidden costs are a common complaint in real car-sharing apps; reviewers of Getaround, for example, criticise fees that only appear at checkout (App Store, Getaround). Fake Uber's approach of showing the full cost early avoids that frustration.

The screen ends with a fixed bar holding Back and **Save · 165 credits**. Putting the amount on the button means the label states the consequence of pressing it. Because the bar stays fixed while the long form scrolls, Save is always within thumb reach (Hoober, 2013).

Back is a plain text action on the dark bar, which makes cancelling easy without competing with the orange Save button.

7. Driver documents

The *Driver details* card is the app's main piece of extra functionality. Instead of typing a name and licence number, the renter adds two photos: a **Driver ID Photo** and a **Driver Licence Photo**. Each is a large square tile with an upload icon and the hint "Tap to upload". Tapping a tile opens the phone's gallery; once a photo is chosen, the tile shows a thumbnail, a green tick and "Tap to change". If the user presses Save without both photos, both tiles turn red and say "Required". Three clearly different states (empty, added and missing) mean the user always knows what is still needed, and the hint text turns an empty tile into an instruction, as Kaplan's empty-state guidelines recommend (Kaplan, 2021).

The justification is trust. A real rental company must know who is driving its car, and the driving licence is the document that legally matters for a car. Collecting the photos inside the rent flow, next to the cost, makes the requirement part of the same decision instead of a surprise later.

The gallery is opened with Android's system **photo picker** (`ActivityResultContracts.PickVisualMedia`) rather than by asking for storage access. Google's Android team describes the picker as "private by design": users choose specific images "without needing to grant access to their entire media library", and the picker was backported to Android 11 and 12 through a Google Play system update (Rezgui, 2022). For a proof-of-concept this is also simpler, because no permission dialog interrupts the flow. The app keeps only a reference (a `Uri`) to each photo in memory, in line with the brief's rule against persistent data.

Two limits must be stated honestly. First, choosing a photo from a gallery does not prove the renter is a real person or that the documents are theirs. Production apps verify identity with electronic know-your-customer checks such as reading the document text, matching a live selfie to the photo, or reading the chip in a modern ID card. The feature here demonstrates the interface for collecting documents, not real verification. Second, ID documents are personal data. Vietnam's Decree 13/2023/NĐ-CP lists ID numbers as basic personal data and states that consent must be clear and specific, and that silence does not count as consent (LuatVietnam, 2023). A future version should add a short consent line and checkbox beside the tiles and explain that photos are used only for this booking.

8. Errors and feedback

The brief requires error checking that stops the user from returning to Home until the problem is fixed or the booking is cancelled. Fake Uber meets this with errors that appear where the problem is, say what is wrong in numbers, and offer a way out. When a 6-day rental of a 90-credit car is attempted, the limit bar turns red, the balance after rental shows "-40 cr" in red, a message inside the Credits card reads *This rental costs 540 credits. One rental can't go over 400. Pick fewer days or press Back to cancel.*, and the missing document tiles turn red. A short-balance case is just as specific: *You have 140, this rental needs 180.*

These messages follow almost every point in Nielsen Norman Group's error-message guidelines: show the error close to its source, use noticeable styling without relying on colour alone, describe the specific problem in plain language, offer a constructive remedy, keep the tone non-judgemental, and preserve the user's input (Neusesser and Sunwall, 2023). Each error pairs red with an icon, text or a minus sign, and nothing the user has chosen is cleared. The Save button also stays enabled in the error state. That is a good decision: Vitaly Friedman argues that disabled buttons leave users guessing what is wrong (Friedman, 2021), whereas pressing an enabled Save here explains every problem at once.

All notifications now use a dark Snackbar at the bottom of the screen. Earlier drafts placed white banners at the top, but that cannot be built as a standard Toast: for apps targeting Android 11, `setGravity()` has no effect on text toasts (Künneth, 2020). The Material Components documentation describes Snackbars as brief messages at the bottom of the screen that do not interrupt the user, are read out automatically by TalkBack, can carry one action, and can be lifted above a bottom bar with `setAnchorView` (Material Components for Android, Snackbar). On the Rent screen, *Can't save yet* sits directly above the Back/Save bar for this reason.

The final design also separates two outcomes that earlier versions confused. Pressing Back on the Rent screen shows **Rental not saved – Car Model was not rented. No credits used.** Cancelling an existing booking from *My rentals* shows **Booking cancelled · 165 credits refunded – Car Model is back in the list.** The second message names the refund and the car's return, which is exactly what the brief asks the user to see. Because cancelling a booking is quick to do by mistake, adding an **Undo** action to that Snackbar would be a cheap safety net and supports Nielsen's third heuristic, user control and freedom.

9. Dark mode

The brief requires a Switch at the top of the page that changes the theme on at least two screens. Fake Uber puts a sun/moon switch beside the credit balance on both Home and Rent, and every colour token has a dark value. The dark theme follows the principles in Google's Material dark-theme codelab: a dark grey background (#111419) rather than pure black, surfaces that get lighter as they come forward (background, then cards at #1B1F26, then chips at #262B33), and off-white text (#F4F1EB) instead of pure white (Google Codelabs, Design a dark theme). Elements that were black in light mode, such as the credit pill, the Next button and the bottom bar, invert to cream so they keep their prominence.

Nielsen Norman Group's study of dark mode lists the problems designers most often miss: thin dividers and cards that vanish, saturated colours that barely meet contrast thresholds, and shadows that no longer show depth (Kohler and Zhang, 2023). Fake Uber handles most of these. Cards are separated by colour steps rather than shadows, and measured contrast is strong: 16.4:1 for main text, 6.8:1 for secondary text on cards, and 6.6:1 for orange on the background. The one failure is the error red (#E11D2E), which reaches only 3.5:1 on dark cards; a lighter red such as #F87171 for dark mode would fix it.

The same study found that users think of dark mode as a system setting and are annoyed when an app ignores it. The in-app switch is required by the brief, so the best compromise is the one Android's documentation recommends: start in *follow system* mode and let the switch override it (Android Developers, Dark theme). The same documentation notes that `setDefaultNightMode()` recreates running activities. For this app that means the balance, favourites and bookings must live in a ViewModel, or pressing the switch would wipe the user's rental.

10. Accessibility

Accessibility is where the design is weakest, but every problem found can be fixed by changing a colour value or a minimum size, without moving anything on screen. Contrast was measured against WCAG 2 success criterion 1.4.3, which requires 4.5:1 for normal text and 3:1 for large text (W3C, Understanding 1.4.3), and 1.4.11, which requires 3:1 for the visual boundaries and states of controls (W3C, Understanding 1.4.11). The core pairs pass comfortably: main text on the cream background reaches 15.8:1, dark text on the orange buttons reaches 6.4:1, and white text on the dark Snackbar reaches about 13.6:1.

Six light-theme pairs and one dark-theme pair fail. Most are fixed by changing a single colour token, so one edit repairs every screen that uses it. The orange fix keeps dark text on orange buttons passing at 5.04:1.

Where it appears	Colours (text / background)	Measured	Needs	Fix
"drive today." headline (large text)	#F97316 / #F4F0E8	2.47:1	3:1	#EA580C → 3.13:1
Slider thumb, favourite heart	#F97316 / #FCFBF8	2.71:1	3:1	#EA580C → 3.44:1
"Tap to upload" on document tiles	#716C65 / #EFEAE2	4.35:1	4.5:1	#625D57 → 5.44:1
Stat chip labels (Year, Kilometres)	#716C65 / #ECE6DC	4.19:1	4.5:1	#625D57 → 5.25:1
Text inside the pink error box	#E11D2E / #FDE3E3	3.91:1	4.5:1	#B91C1C → 5.32:1
"RENTED · PICK UP NOW" label	#2F9E5E / #FCFBF8	3.29:1	4.5:1	#1F7A45 → 5.17:1
Error text, dark theme	#E11D2E / #1B1F26	3.48:1	4.5:1	#F87171 → 5.98:1

Touch targets are the second issue. Android recommends at least 48 × 48dp for anything tappable (Android Developers, Make apps more accessible). Rent and Next (52dp) and Save (48dp) meet it; five controls do not. None of them needs to look bigger: a 48dp minimum width and height, or a Material component that already reserves a 48dp touch area, fixes each one invisibly.

Control	Drawn height	Fix when building
Favourite heart	about 42dp	48dp touch area around the 42dp circle
Back button (Rent)	about 42dp	Icon button with a 48dp target
Dark-mode switch	about 36dp	`MaterialSwitch`, which keeps its own 48dp target
Cancel booking	36dp	Raise the button to 48dp
Search clear (×)	24dp	`TextInputLayout` clear-text end icon (48dp)

Several things already pass. States never rely on colour alone: the favourite heart changes shape, errors add an icon and words, and a negative balance shows a minus sign. The designer's own accessibility notes also commit to screen-reader labels for every icon-only control, including "Upload driver ID photo" and "Upload driver licence photo" for the

document tiles, and to the bottom Snackbar. When the app is built, text sizes should use `sp` so they scale with the user's font setting (CommonsWare, 2012), and Google's free Accessibility Scanner should be run on every screen to confirm the fixes.

11. Limitations, future work and conclusion

This evaluation has three limits. First, it is an expert review, not a user study. Heuristic evaluation is good at finding likely problems, but it cannot show how often real renters stumble or how long tasks take. Second, the final screens exist as high-fidelity sketches, so contrast figures come from the design's colour tokens and touch-target sizes from the drawn components; both should be re-checked on a device once the app is built. Third, the document upload shows how identity documents could be collected, not whether they are genuine, as section 7 explained.

The most valuable next steps are small. A short think-aloud test with a handful of classmates renting one car would show whether the one-car-at-a-time layout and the Credits card work as well as the evidence suggests. The colour and touch-target fixes in section 10 should be applied in code from the start. Four interface additions would close the remaining gaps: a "Sorted by" line on Home, an Undo action on the booking-cancelled Snackbar, a consent line beside the document tiles, and real, freely licensed car photos to replace the placeholders, as the brief requires.

The heuristic evaluation behind this report is summarised below. Six heuristics are met strongly, two are met well with a small gap that has a one-line fix, and two are met adequately for an app this small.

Nielsen heuristic	Evidence in the final design	Verdict
1. Visibility of system status	Credit balance on every screen, "1 / 5" counter, live Credits card	Strong; add a "Sorted by" line
2. Match with the real world	"Pick it up now", "credits / day", "Return within a week"	Strong
3. User control and freedom	Back cancels, Cancel booking, Clear search	Good; add Undo to the cancel Snackbar
4. Consistency and standards	Same components on all 14 screens; standard bottom Snackbar	Strong
5. Error prevention	Discrete 1–7 slider, live balance and limit bar before Save	Strong

Nielsen heuristic	Evidence in the final design	Verdict
6. Recognition rather than recall	Car details restated on Rent, labelled sort options, favourite chips	Strong
7. Flexibility and efficiency	Long-press to favourite, tap a chip to jump to a car	Adequate for a five-car app
8. Aesthetic and minimalist design	One orange action per screen, titled blocks	Good; price shown twice on the car card
9. Recognise, diagnose, recover from errors	Specific numbers plus a remedy in every message	Strong
10. Help and documentation	Helper lines and empty-state hints	Adequate; no separate help needed

In conclusion, the final Fake Uber design does what a proof-of-concept rental app needs to do. It covers every requirement in the brief, and most of its decisions are supported by published evidence rather than taste. The labelled "1 / 5" counter makes a weak carousel pattern usable. The live Credits card prevents errors instead of reporting them. Errors and Snackbars say exactly what happened and what to do next. Empty states teach the user what to do. Its weaknesses are concentrated in colour contrast and target size, and both can be fixed without redesigning a single screen.

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